THE UK UNIVERSITY INTEGRATION BEE 2025/26

$$\int$$

ROUND ONE MARK SCHEME

1. $\tan^{-1}\left(\frac{\pi}{4}\right)$
2. 0
3. $\frac{\pi}{\sqrt{2}}$
4. ai

   Accept other equivalent forms.
5. 1
6. $\frac{\sqrt{\pi}}{2}$

   $\Gamma\left(\frac{3}{2}\right)$

   $\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$

   Half: $\sqrt{\pi}$ or equivalent.
7. 1
8. 1
9. 2
10. $\frac{\pi}{2}\sinh^{-1}(b)$

    $\frac{\pi}{2}\ln\left(b+\sqrt{b^2+1}\right)$
11. $\ln\left|\sin^{-1}(\ln(x))\right|+C$
12. $1-\ln(2)$

    Half: Correct answer with alternating harmonic series but not explicit value.
13. $\frac{4}{3}$
14. 0
15. $2\sqrt{2}$
16. $\frac{3\pi}{8\sqrt{2}}$
17. $\frac{1}{1-n}\ln\left(\frac{n}{n+1}\right)$
18. $\frac{\pi}{2}\ln(2)$

    Half: Wrong sign or off by a factor of exactly 2 or both.
19. 0
20. $\frac{\pi}{4}\left(e^2-1\right)$

21. $\dfrac{2025}{2026}$

    Half: $\dfrac{2024}{2025}$

22. $\dfrac{1}{\ln(2)}$

23. e

24. 0

25. $\dfrac{675}{2}$

26. $\ln\left(\dfrac{4}{\pi}\right)$

27. $\dfrac{(\sqrt{3}-1)^x - \ln\left(1+(\sqrt{3}-1)^x\right)}{\ln\left(\sqrt{3}-1\right)} + C$

28. $90(45-\sqrt{2024}) - 2\ln\left(45-\sqrt{2024}\right) - 3\ln(2)$

29. 812

30. $\dfrac{3\pi\ln(2)}{8}$

    $\dfrac{\ln(2)(\tan^{-1}(3)+\tan^{-1}(2))}{2}$